

Modus ponens is a fundamental inference rule in propositional logic. If we know p is true and p implies q , we can conclude q is true.

$$\frac{\begin{array}{l} p \\ p \Rightarrow q \end{array}}{q} \quad \text{modus ponens}$$

The horizontal line (generated by `—`) separates premises from conclusion. The label in brackets appears to the right of the conclusion.

Here's a more complex example with multiple premises:

$$\frac{\begin{array}{l} p \Rightarrow q \\ q \Rightarrow r \\ p \end{array}}{r} \quad \text{chain of implications}$$

We can also use `INFRULE` without labels:

$$\frac{A \wedge B}{A} \quad \text{conjunction elimination}$$